Introduction to Mathematics for Deep Learning: Homeworks #4 Solutions

1. If $f : \mathbb{R}^d \to \mathbb{R}$ is twice-differentiable, show that the function f is convex if and only if

$$\nabla^2 f \succeq 0.$$

In other words, $\nabla^2 f(x)$ is positive semi-definite for every x, and equivalently, $v^T \nabla^2 f(x) v \geq 0$ for all vector $v \in \mathbb{R}^d$.

Solution:
For notational simplicity, let $H_f(x) = \nabla^2 f(x)$.

(1) If f is convex, then $H_f(x)$ is positive semi-definite
By definition, f is convex if for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y).$$

Equivalently, f is convex if and only if for any $x \in \mathbb{R}^n$, the following inequality holds:

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall y \in \mathbb{R}^n.$$

Using the second-order Taylor expansion of $f(y)$ around x, we have:

$$f(y) = f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}(y - x)^T H_f(x)(y - x) + o(\|y - x\|^2)$$

where $o(\|y - x\|^2)$ is a negligible term.

For f to be convex, the quadratic term must satisfy:

$$\frac{1}{2}(y - x)^T H_f(x)(y - x) \geq 0, \quad \forall y \in \mathbb{R}^n.$$

Let $z = y - x$. Then:

$$z^T H_f(x) z \geq 0, \quad \forall z \in \mathbb{R}^n.$$

This implies that $H_f(x)$ is positive semi-definite.

(2) If $H_f(x)$ is positive semi-definite, then f is convex
Assume that $H_f(x)$ is positive semi-definite for all $x \in \mathbb{R}^n$. For any $x, y \in \mathbb{R}^n$, consider the function $g(\lambda)$ defined as:

$$g(\lambda) = f((1-\lambda)x + \lambda y), \quad \lambda \in [0, 1].$$

By the chain rule, the first derivative of $g(\lambda)$ is:

$$g'(\lambda) = \nabla f((1-\lambda)x + \lambda y)^T (y - x).$$

The second derivative of $g(\lambda)$ is:

$$g''(\lambda) = (y-x)^T H_f((1-\lambda)x + \lambda y)(y-x).$$

Since H_f is positive semi-definite, we have:

$$g''(\lambda) \geq 0, \quad \forall \lambda \in [0,1].$$

Thus, $g(\lambda)$ is a convex function of λ. This implies:

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y), \quad \forall \lambda \in [0,1].$$

Therefore, f is convex.

2. Consider gradient descent for a convex quadratic function:

$$f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x} - \mathbf{b}^T\mathbf{x},$$

where $\mathbf{Q} \in \mathbb{R}^{n\times n}$ is a positive definite matrix (the Hessian), and $\mathbf{b} \in \mathbb{R}^n$. The minimum is at $\mathbf{x}^* = \mathbf{Q}^{-1}\mathbf{b}$.

The update rule for gradient descent is:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta\nabla f(\mathbf{x}_t),$$

where $\nabla f(\mathbf{x}_t) = \mathbf{Q}\mathbf{x}_t - \mathbf{b}$. Suppose the largest eigenvalue of $\mathbf{Q}$ is L, and $0 < \eta < 2/L$. Let $\lambda_{\min}$ be the smallest eigenvalue of $\mathbf{Q}$. Show that the error $\mathbf{e}_t = \mathbf{x}_t - \mathbf{x}^*$ converges in $\mathcal{O}(c^t)$. Find c (in terms of the eigenvalues of $\mathbf{Q}$ and η).

Solution:
Substituting the gradient into the update:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta(\mathbf{Q}\mathbf{x}_t - \mathbf{b}).$$

Rearranging gives

$$\mathbf{x}_{t+1} = (\mathbf{I} - \eta\mathbf{Q})\mathbf{x}_t + \eta\mathbf{b}.$$

Let $\mathbf{x}^* = \mathbf{Q}^{-1}\mathbf{b}$ be the minimizer, and define the error

$$\mathbf{e}_t = \mathbf{x}_t - \mathbf{x}^*.$$

Then

$$\mathbf{e}_{t+1} = \mathbf{x}_{t+1} - \mathbf{x}^* = (\mathbf{I} - \eta\mathbf{Q})\mathbf{x}_t + \eta\mathbf{b} - \mathbf{x}^*.$$

Using $\mathbf{Q}\mathbf{x}^* = \mathbf{b}$, we have

$$\mathbf{x}^* = (\mathbf{I} - \eta\mathbf{Q})\mathbf{x}^* + \eta\mathbf{b},$$

so subtracting this relation gives

$$\mathbf{e}_{t+1} = (\mathbf{I} - \eta\mathbf{Q})(\mathbf{x}_t - \mathbf{x}^*) = (\mathbf{I} - \eta\mathbf{Q})\mathbf{e}_t.$$

Iterating,

$$\mathbf{e}_t = (\mathbf{I} - \eta\mathbf{Q})^t\mathbf{e}_0.$$

The convergence rate is governed by the spectral radius of the matrix

$$\mathbf{M} := \mathbf{I} - \eta\mathbf{Q}.$$

Eigenvalue analysis. Since $\mathbf{Q}$ is symmetric positive definite, it admits an eigendecomposition

$$\mathbf{Q} = \mathbf{V}\Lambda\mathbf{V}^\top,$$

where $\mathbf{V}$ is orthogonal and

$$\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n), \quad 0 < \lambda_{\min} = \lambda_1 \le \cdots \le \lambda_n = L.$$

Then

$$\mathbf{M} = \mathbf{I} - \eta\mathbf{Q} = \mathbf{V}(\mathbf{I} - \eta\Lambda)\mathbf{V}^\top,$$

so the eigenvalues of $\mathbf{M}$ are

$$\mu_i = 1 - \eta\lambda_i, \quad i = 1, \dots, n.$$

Hence the spectral radius is

$$\rho(\mathbf{M}) = \max_i |\mu_i| = \max_i |1 - \eta\lambda_i|.$$

Because all $\lambda_i \in [\lambda_{\min}, L]$ and $0 < \eta < 2/L$, we have

$$-1 < 1 - \eta\lambda_i < 1 \quad \text{for all } i,$$

so $\rho(\mathbf{M}) < 1$ and gradient descent converges linearly.

Moreover, the function $g(\lambda) = |1 - \eta\lambda|$ on the interval $[\lambda_{\min}, L]$ attains its maximum at one of the endpoints:

$$\rho(\mathbf{M}) = \max_{\lambda \in [\lambda_{\min}, L]} |1 - \eta\lambda| = \max\left\{|1 - \eta\lambda_{\min}|,\ |1 - \eta L|\right\} =: c.$$

Thus

$$\|\mathbf{e}_t\|_2 = \|\mathbf{M}^t\mathbf{e}_0\|_2 \le \|\mathbf{M}\|_2^t \, \|\mathbf{e}_0\|_2 \le c^t \|\mathbf{e}_0\|_2,$$

with

$$c = \max\left\{|1 - \eta\lambda_{\min}|,\ |1 - \eta L|\right\} < 1.$$

This shows that

$$\|\mathbf{e}_t\|_2 = \mathcal{O}(c^t).$$

Remark (role of $\lambda_{\min}$). If we additionally choose a "safe" step size $0 < \eta \le 1/L$, then $1 - \eta\lambda$ is *decreasing* in λ on $[\lambda_{\min}, L]$ and remains nonnegative. In that case,

$$c = \max_{\lambda \in [\lambda_{\min}, L]} (1 - \eta\lambda) = 1 - \eta\lambda_{\min},$$

which is the commonly quoted rate

$$\|\mathbf{e}_t\|_2 = \mathcal{O}\big((1 - \eta\lambda_{\min})^t\big).$$

3. Consider the ridge regression problem where the model $f_\theta(x) = \theta^T x$ is linear, and data $\{(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)})\}$ are given.

   (a) As we have seen in the class, show that the empirical risk can be expressed in matrix formula

   $$\mathcal{L}(\theta) = \|X\theta - Y\|^2.$$

   (b) The objective of ridge regression is to minimize

   $$\mathcal{L}(\theta) + \lambda\|\theta\|^2 = \|X\theta - Y\|^2 + \lambda\|\theta\|^2.$$

   for some $\lambda > 0$. Suppose X is a full rank matrix. Show that the optimal solution is

   $$\theta^\star = (X^T X + \lambda I)^{-1} X^T Y.$$

Solution:

$$\|X\theta - Y\|^2 + \lambda\|\theta\|^2 = \left\| \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \theta - \begin{bmatrix} Y \\ 0 \end{bmatrix} \right\|^2 \tag{1}$$

$$\tag{2}$$

Thus, the minimum achieving θ is

$$\theta^\star = \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^\dagger \begin{bmatrix} Y \\ 0 \end{bmatrix} \tag{3}$$

$$= \left(\begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^T \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \right)^{-1} \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix}^T \begin{bmatrix} Y \\ 0 \end{bmatrix} \tag{4}$$

$$= \left(\begin{bmatrix} X^T & \sqrt{\lambda} I \end{bmatrix} \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \right)^{-1} \begin{bmatrix} X^T & \sqrt{\lambda} I \end{bmatrix} \begin{bmatrix} Y \\ 0 \end{bmatrix} \tag{5}$$

$$= \left(X^T X + \lambda I\right)^{-1} X^T Y. \tag{6}$$

4. For μ-strongly convex function f, show that

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y) - \frac{\mu}{2}\theta(1-\theta)\|x - y\|^2$$

for all x, y.

Solution:
Let $z = \theta x + (1-\theta)y$. Then,

$$f(x) \geq f(z) + \langle \nabla f(z), x - z \rangle + \frac{\mu}{2}\|x - z\|^2$$

$$= f(z) + (1-\theta)\langle \nabla f(z), x - y\rangle + \frac{\mu}{2}(1-\theta)^2\|x-y\|^2$$

Similarly,

$$\begin{aligned} f(y) \geq & f(z) + \langle \nabla f(z), y - z\rangle + \frac{\mu}{2}\|y-z\|^2 \\ = & f(z) - \theta\langle \nabla f(z), x - y\rangle + \frac{\mu}{2}\theta^2\|x-y\|^2 \end{aligned}$$

Then,

$$\begin{aligned} \theta f(x) + (1-\theta) f(y) \geq & f(z) + \frac{\mu}{2}\theta(1-\theta)\|x-y\|^2 \\ = & f(\theta x + (1-\theta)y) + \frac{\mu}{2}\theta(1-\theta)\|x-y\|^2 \end{aligned}$$

which concludes the proof.

5. For a L-smooth convex function f and $0 < \theta < 1$, consider the gradient descent problem with slightly changed parametrization.

$$x_{k+1} = x_k - \frac{2\theta}{L}\nabla f(x_k).$$

(a) Using the co-coercivity

$$\langle \nabla f(y) - \nabla f(x), y - x\rangle \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2,$$

show that

$$\|x_{k+1} - x_k\|^2 \leq \|x_k - x_{k-1}\|^2.$$

(b) Let

$$V_k = k\frac{1-\theta}{\theta}\|x_k - x_{k-1}\|^2 + \|x_k - x^\star\|^2$$

where $x^\star$ is the global minimum. Show that $V_{k+1} \leq V_k$.

(c) Show that

$$\|x_{k+1} - x_k\|^2 \leq \frac{\theta}{1-\theta}\frac{1}{k+1}\|x_0 - x^\star\|^2$$

Solution:

(a) We have

$$\begin{aligned}\|x_{k+1}-x_k\|^2 =&\|x_k-x_{k-1}-\frac{2\theta}{L}\nabla(f(x_k)-\nabla f(x_{k-1}))\|^2\\ =&\|x_k-x_{k-1}\|^2\\ &-\frac{4\theta}{L}\left(\langle\nabla f(x_k)-\nabla f(x_{k-1}),x_k-x_{k-1}\rangle-\frac{\theta}{L}\|\nabla f(x_k)-\nabla f(x_{k-1})\|^2\right)\\ \leq&\|x_k-x_{k-1}\|^2\\ &-\frac{4\theta}{L}\left(\langle\nabla f(x_k)-\nabla f(x_{k-1}),x_k-x_{k-1}\rangle-\frac{1}{L}\|\nabla f(x_k)-\nabla f(x_{k-1})\|^2\right)\\ \leq&\|x_k-x_{k-1}\|^2.\end{aligned}$$

(b) We have

$$\begin{aligned}V_{k+1}-V_k =&(k+1)\frac{1-\theta}{\theta}\|x_{k+1}-x_k\|^2+\|x_{k+1}-x^\star\|^2\\ &-k\frac{1-\theta}{\theta}\|x_k-x_{k-1}\|^2-\|x_k-x^\star\|^2\\ \leq&\frac{1-\theta}{\theta}\|x_{k+1}-x_k\|^2+\|x_{k+1}-x^\star\|^2-\|x_k-x^\star\|^2\\ =&\frac{1-\theta}{\theta}\frac{4\theta^2}{L^2}\|\nabla f(x_k)\|^2+\|x_k-x^\star-\frac{2\theta}{L}\nabla f(x_k)\|^2-\|x_k-x^\star\|^2\\ =&\frac{1-\theta}{\theta}\frac{4\theta^2}{L^2}\|\nabla f(x_k)\|^2+\frac{4\theta^2}{L^2}\|\nabla f(x_k)\|^2-\frac{4\theta}{L}\langle x_k-x^\star,\nabla f(x_k)\rangle\\ =&\frac{4\theta}{L}\left(\frac{1}{L}\|\nabla f(x_k)\|^2-\langle x_k-x^\star,\nabla f(x_k)\rangle\right)\\ =&\frac{4\theta}{L}\left(\frac{1}{L}\|\nabla f(x_k)-\nabla f(x^\star)\|^2-\langle x_k-x^\star,\nabla f(x_k)-\nabla f(x^\star)\rangle\right)\\ \leq&0.\end{aligned}$$

where the last inequality is co-coercivity again with $\nabla f(x^\star)=0$.

(c) Again,

$$\begin{aligned}(k+1)\frac{1-\theta}{\theta}\|x_{k+1}-x_k\|^2 \leq&V_{k+1}\\ \leq&V_0\\ \leq&\|x_0-x^\star\|^2.\end{aligned}$$

Thus,

$$\|x_{k+1}-x_k\|^2\leq\frac{\theta}{1-\theta}\frac{1}{k+1}\|x_0-x^\star\|^2$$

6. Let $A\in\mathbb{R}^{n\times n}$ be an orthogonal matrix, i.e. $A^\top A=I$. Show that all singular values of A are equal to 1. [Hint: Recall that the singular values of A are defined as the (nonnegative) square roots of the eigenvalues of $A^\top A$.]

Solution:
Start from the SVD $A = U\Sigma V^\top$. Then

$$A^\top A = (U\Sigma V^\top)^\top (U\Sigma V^\top) = V\Sigma^\top U^\top U\Sigma V^\top.$$

Since U is orthogonal, $U^\top U = I$, and since Σ is diagonal, $\Sigma^\top = \Sigma$. Hence

$$A^\top A = V\Sigma^2 V^\top.$$

On the other hand, since A is orthogonal, we are given that

$$A^\top A = I.$$

Combining this with the expression from part (a), we obtain

$$V\Sigma^2 V^\top = I.$$

Multiply on the left by $V^\top$ and on the right by V:

$$\Sigma^2 = V^\top I V = I.$$

Thus $\Sigma^2 = I$.

The matrix Σ is diagonal with nonnegative entries $\sigma_1, \ldots, \sigma_n$ on the diagonal, and $\Sigma^2 = I$ means

$$\sigma_i^2 = 1 \quad \text{for all } i.$$

Since by definition singular values are nonnegative, we must have

$$\sigma_i = 1 \quad \text{for all } i = 1, \ldots, n.$$

Therefore all singular values of the orthogonal matrix A are equal to 1.

7. Let $A \in \mathbb{R}^{n\times n}$ be an orthogonal matrix, i.e. $A^\top A = I$. Consider any rank-k matrix B (obtained by any approximation method).

   (a) Show that for every rank-k matrix B,

   $$\|A - B\|_2 \;\geq\; 1.$$

   (b) Show that this lower bound is tight, i.e., there exists at least one rank-k matrix B such that

   $$\|A - B\|_2 = 1.$$

   Find the simplest one.

Solution:

(a) Since A is orthogonal, we have $A^\top A = I$. The eigenvalues of I are all equal to 1, so the singular values of A are

$$\sigma_1(A) = \sigma_2(A) = \cdots = \sigma_n(A) = 1.$$

By the Eckart–Young–Mirsky theorem, for any integer $k < n$,

$$\min_{\text{rank}(B) \le k} \|A - B\|_2 = \sigma_{k+1}(A).$$

Since $\sigma_{k+1}(A) = 1$ for an orthogonal matrix, we obtain

$$\min_{\text{rank}(B) \le k} \|A - B\|_2 = 1.$$

By definition of the minimum, this implies that for *every* rank-k matrix B,

$$\|A - B\|_2 \ \ge \ 1.$$

(b) For tightness, it suffices to exhibit a specific rank-k matrix B with error exactly 1.

One simple choice is $B = 0$, which has rank $0 \le k$. Then

$$\|A - 0\|_2 = \|A\|_2 = \sigma_1(A) = 1.$$

Thus the lower bound 1 is attained.

Alternatively, one could take B to be the rank-k truncated SVD of A; since all singular values are equal to 1, its $(k+1)$-st singular value is still 1, and Eckart–Young–Mirsky again implies

$$\|A - B\|_2 = \sigma_{k+1}(A) = 1.$$

Therefore, for orthogonal A, any rank-k approximation B satisfies $\|A - B\|_2 \ge 1$, and the bound is tight.